Boston Housing Predictor

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1 Boston Suburbs Housing Price Predictor

This report investigates the key drivers of median home values across Boston suburbs using multiple linear regression. The dataset contains 400 suburbs and 12 predictor variables, ranging from socioeconomic indicators such as the proportion of lower-status residents (`lstat`) and crime rate (`crim`), to structural characteristics such as average room count (`rm`) and proximity to employment centres (`dis`). The analysis follows a structured modelling pipeline: exploratory data analysis, full linear regression, backward stepwise selection using AIC and BIC criteria, and Ridge and LASSO regularisation. Model performance is assessed using 10-fold cross-validation RMSE and test-set evaluation on a held-out sample. The goal is to identify the most parsimonious and generalisable model that accurately predicts `medv` — the median value of owner-occupied homes in $1,000s.

1.1 Exploration

The first step is to understand the size, structure, missingness, distributions, and relationships in the training data. The response variable is `medv`, the median value of owner-occupied homes in $1000s.

A brief description of the predictors is given in Table 1.

Table 1: Descriptions of predictors in the Boston housing dataset.

Variable	Description
crim	Per capita crime rate by town.
zn	Proportion of residential land zoned for lots over 25,000 sq.ft.
indus	Proportion of non-retail business acres per town.
chas	Charles River dummy variable (= 1 if tract bounds river; 0 otherwise).
nox	Nitrogen oxides concentration (parts per 10 million).
rm	Average number of rooms per dwelling.
age	Proportion of owner-occupied units built prior to 1940.
dis	Weighted mean of distances to five Boston employment centres.
rad	Index of accessibility to radial highways.
tax	Full-value property-tax rate per $10,000.
ptratio	Pupil-teacher ratio by town.
lstat	Lower status of the population (percent).
medv	Median value of owner-occupied homes in $1000s.

A peek into the dataset shows that this training sample has 400 Boston suburbs and 13 variables, of which `chas` is a categorical variable representing suburbs near the Charles River.

The data types of the predictors are shown in Table **??**.

Table 2: **??**Variable types in the Boston housing training data.

Variable	Type
crim	numeric
zn	numeric
indus	numeric
chas	factor
nox	numeric
rm	numeric
age	numeric
dis	numeric
rad	integer
tax	integer
ptratio	numeric
lstat	numeric
medv	numeric

The first 10 rows of the dataset are shown in Table 3.

Table 3: First 10 rows of the Boston housing training data.

crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat	medv
0.61	0	6.20	No	0.51	6.62	80.8	3.27	8	307	17.4	7.60	30.1
0.09	45	3.44	No	0.44	6.95	21.5	6.48	5	398	15.2	5.10	37.0
0.33	0	6.20	No	0.51	8.25	70.4	3.65	8	307	17.4	3.95	48.3
0.29	0	21.89	No	0.62	6.17	93.6	1.61	4	437	21.2	24.16	14.0
9.92	0	18.10	No	0.69	5.85	77.8	1.50	24	666	20.2	29.97	6.3
0.03	0	4.39	No	0.44	6.01	48.5	8.01	3	352	18.8	10.53	17.5
9.19	0	18.10	No	0.70	5.54	100.0	1.58	24	666	20.2	23.60	11.3
0.15	0	10.01	No	0.55	6.02	82.6	2.75	6	432	17.8	10.30	19.2
73.53	0	18.10	No	0.68	5.96	100.0	1.80	24	666	20.2	20.62	8.8
0.03	0	7.07	No	0.47	7.18	61.1	4.97	2	242	17.8	4.03	34.7

1.2 Visualisations

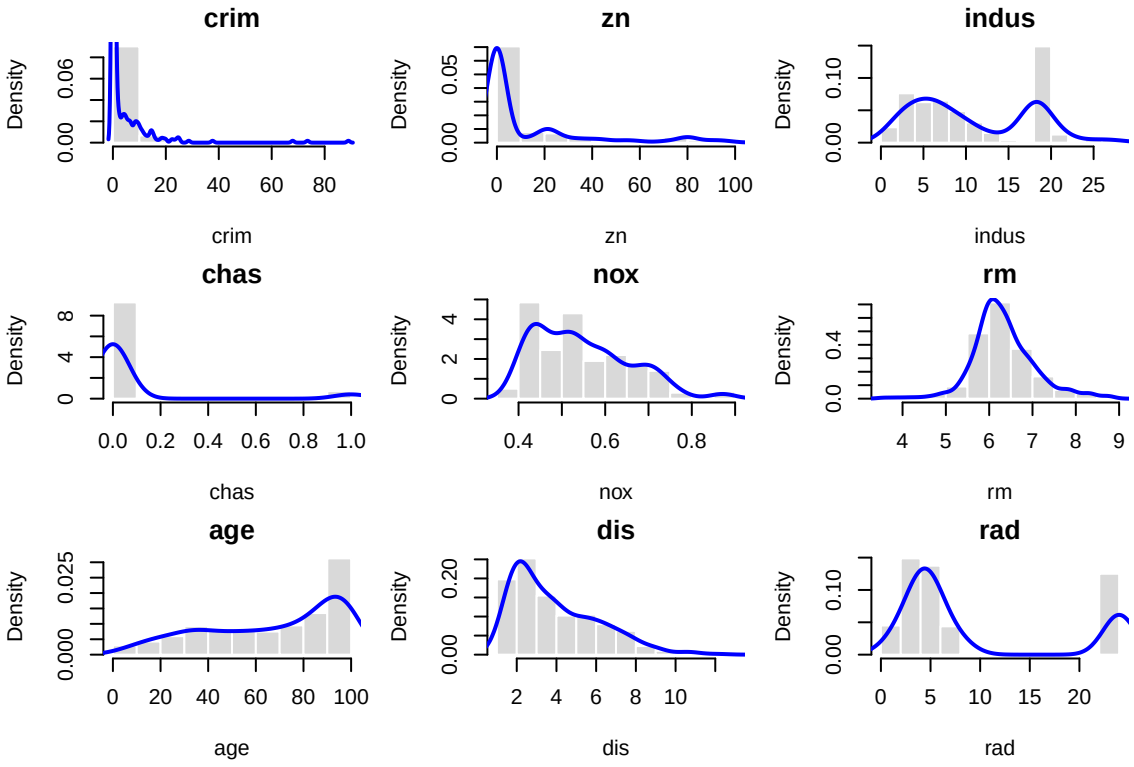

Figure 1: Histograms with density curves for all variables

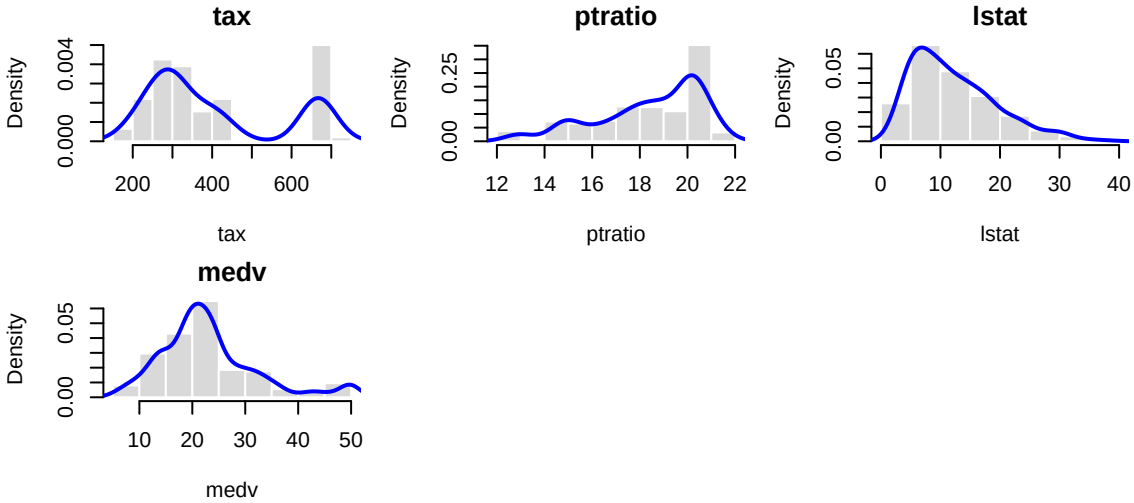

Figure 2: Histograms with density curves for all variables

Findings of interest:

- The dataset has no missing values.
- The `chas` variable is very imbalanced: 92.8% of suburbs do not bound the Charles River, so predictions involving `chas` are based on relatively few observations.
- Several variables are heavily right-skewed, especially `crim`, `zn`, `rad`, `tax`, `ptratio`, and `lstat`, as seen in Figure 2 suggesting these characteristics are concentrated in certain suburbs rather than evenly distributed across Boston.

1.2.1 Explanatory-target relationships

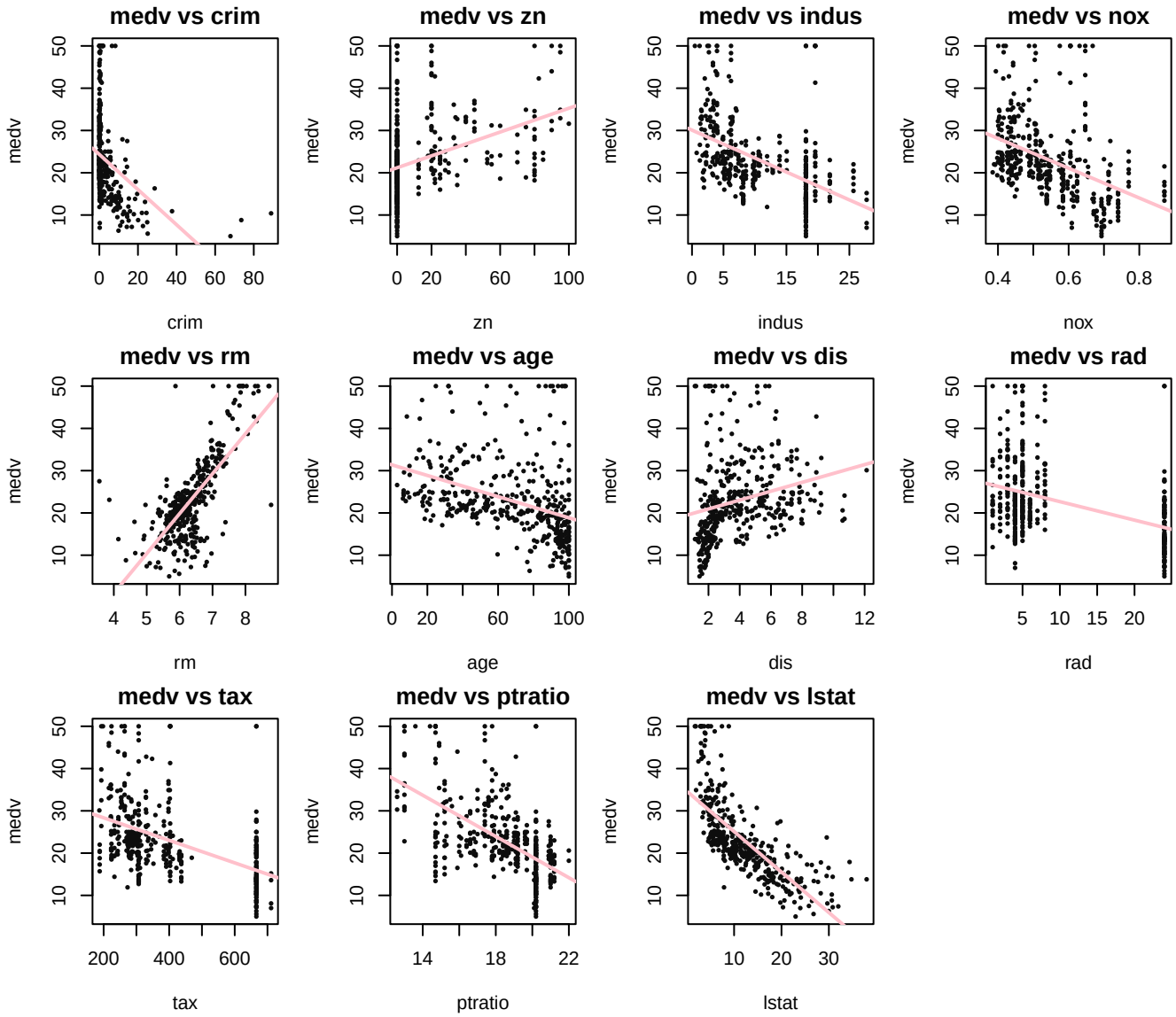

Figure 3: Scatterplots of each explanatory variable against median home value (medv)

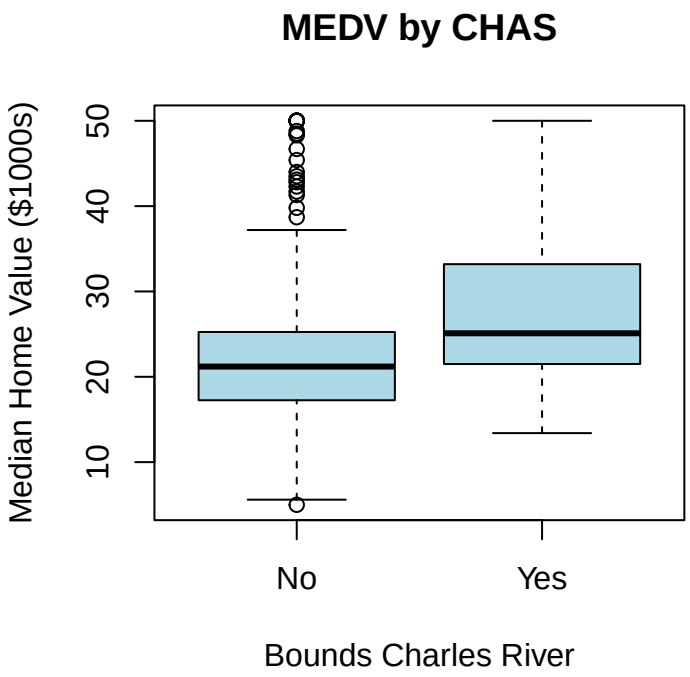

Figure 4: Boxplots of median home value (medv) by Charles River adjacency (chas)

The clearest single-variable relationships with `medv` are with `lstat` (negative), `rm` (positive), and `ptratio` (negative). Median home value tends to rise with more rooms and fall as socioeconomic disadvantage and pupil-teacher ratios increase.

1.2.2 Checking for Outliers

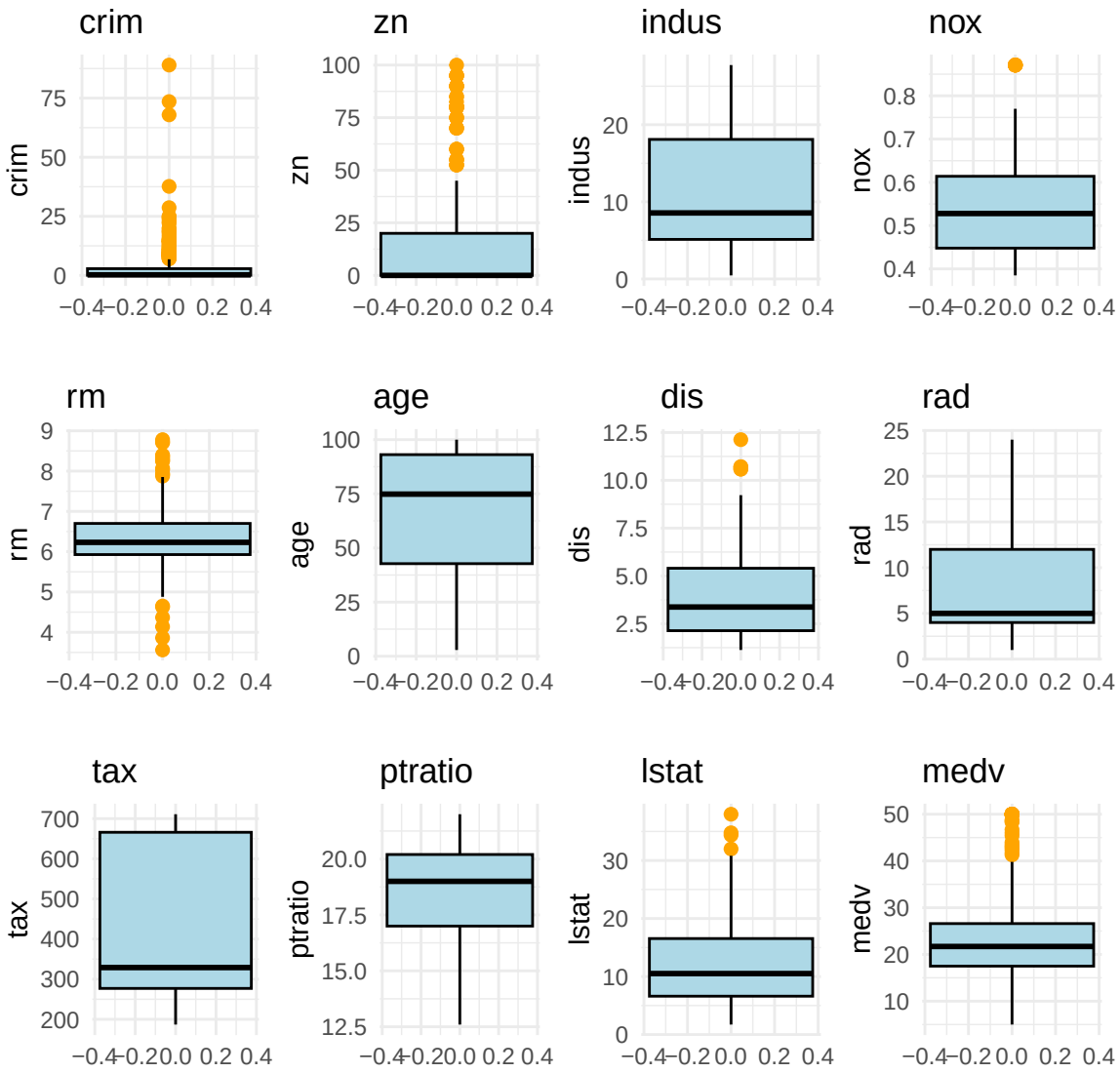

Figure 5: Boxplots for each numeric predictor

The orange points in Figure 5 represent outliers, present in `crim`, `zn`, `nox`, `rm`, `dis`, `lstat`, and `medv`.

1.2.3 Multi-variable Correlation

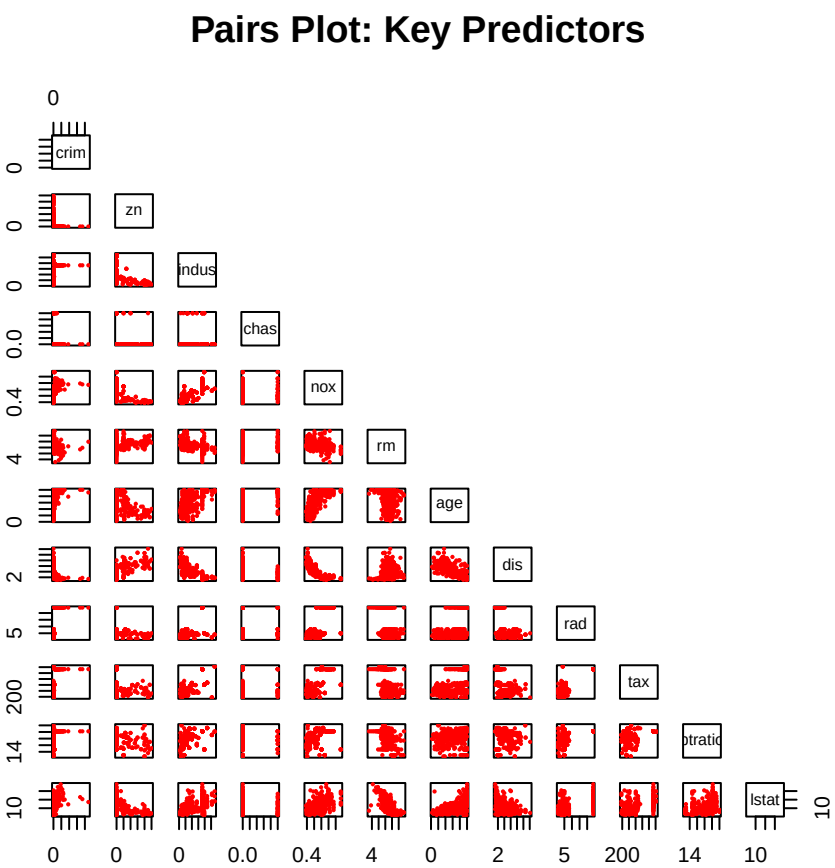

Figure 6: Pairwise scatterplot matrix of the numeric predictors.

The pairwise matrix, (Figure 6) reveals predictors that are strongly related to one another. Notable relationships include `age` and `dis`, as well as `rm` and `lstat`, suggesting these pairs may act as proxies for each other.

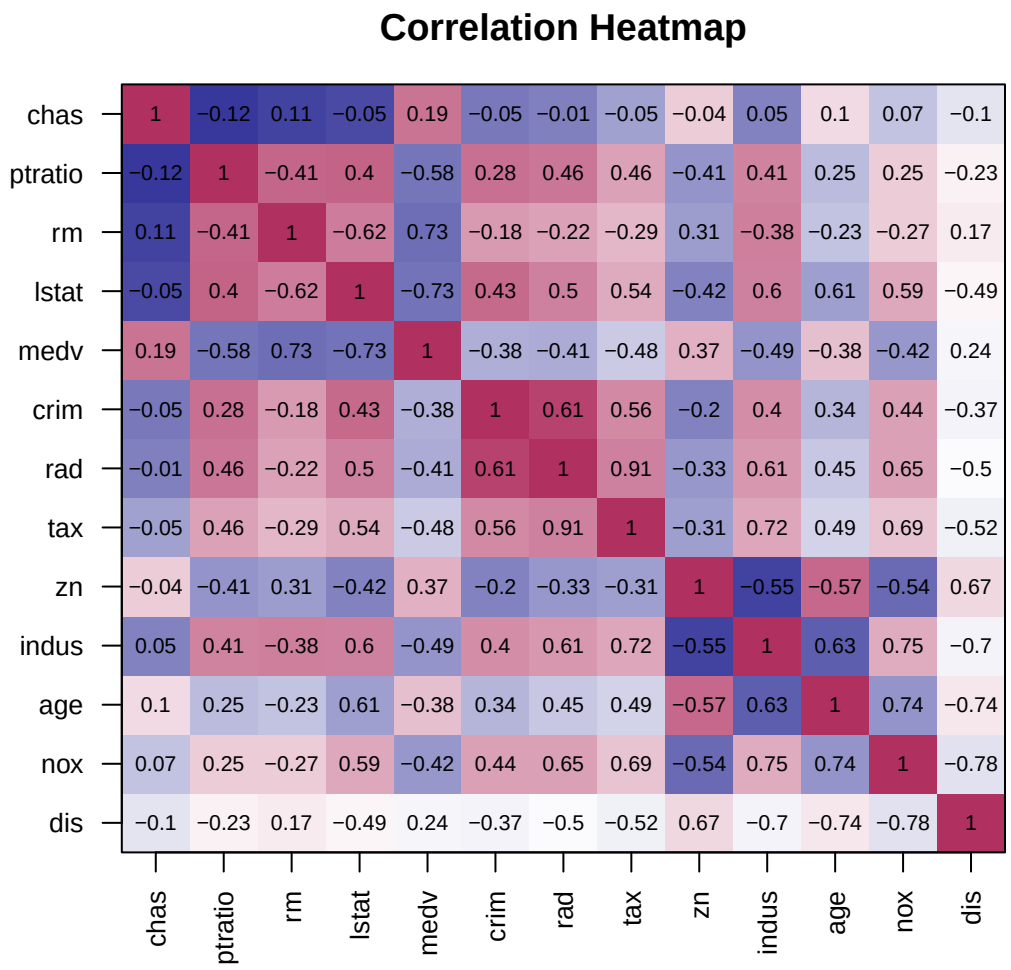

Figure 7: Correlation heatmap

Figure 7 shows the strongest correlations with `medv` are `lstat` (-0.730), `rm` (0.727), `ptratio` (-0.581), `indus` (-0.488), and `tax` (-0.484). Strong correlation among predictors — particularly `rad` and `tax` — suggests potential multicollinearity in a plain linear model. The exploratory phase identifies `lstat`, `rm`, `ptratio`, `tax`, `indus`, `nox`, and `crim` as the most promising predictors.

1.3 Model Building

After setting aside approximately 20% of the data for testing, the saturated model was fitted to the training data.

Table 4: Coefficient estimates, standard errors, t-statistics, and p-values for the full linear regression model

term	estimate	std.error	statistic	p.value
(Intercept)	37.2890	5.3528	6.9663	0.0000
crim	-0.1267	0.0355	-3.5692	0.0004
zn	0.0366	0.0145	2.5171	0.0122
indus	0.0173	0.0651	0.2655	0.7907
chasYes	2.7798	0.9154	3.0367	0.0026
nox	-15.0930	4.4551	-3.3878	0.0008
rm	4.2751	0.4576	9.3427	0.0000
age	-0.0097	0.0143	-0.6734	0.5011
dis	-1.2724	0.2128	-5.9781	0.0000
rad	0.2665	0.0715	3.7263	0.0002
tax	-0.0128	0.0040	-3.2285	0.0014
ptratio	-1.0600	0.1387	-7.6417	0.0000
lstat	-0.4701	0.0567	-8.2974	0.0000

1.3.1 Parameter Significance

From Table 4, `indus` is the most statistically insignificant predictor (p-value = 0.79) — holding all other predictors constant, the proportion of non-retail business acres does very little to impact median sale price. The next least significant predictor is `age` (p-value = 0.50), indicating that the age of housing stock adds negligible explanatory power once other variables are accounted for. Both will be candidates for removal during model improvement.

Table 5: Overall fit statistics for the full linear regression model.

Statistic	Value
Residual standard error	4.5900
Degrees of freedom	387.0000
Multiple R-squared	0.7605
Adjusted R-squared	0.7531
F-statistic	102.4010

The residual standard error of 4.59 indicates that on average, the model's predictions deviate from the true median home value by approximately \$4590 on the training data, with 387 degrees of freedom remaining after estimating all model parameters. The R^2 of 0.7605 means that approximately 76% of the variation in median home values is explained by the model on the training data. The F-statistic of 102.401 tests the null hypothesis that none of the predictors have any linear relationship with `medv`. The extremely small

associated p-value provides overwhelming evidence to reject this null hypothesis, confirming that the model as a whole is statistically significant.

1.3.2 Testing Predictive Fit

Table 6: Test-set performance of the full linear regression model on the held-out Boston suburbs.

Test Statistic	Value
Test MSE	31.8879
Test RMSE	5.6469
Test R-squared	0.5940

The full model explains 59.4% of the variation in median home values on unseen suburbs (Test $R^2 = 0.594$), meaning roughly 40.6% of variation remains unexplained. The Test RMSE of 5.647 means that on average, the model's predictions are off by approximately $5647 from the true value. The Test MSE of 31.888 is the average squared prediction error — because errors are squared, larger individual mistakes are penalised disproportionately, making the MSE sensitive to the extreme home values observed in the right-skewed `medv` distribution. This value serves as the baseline benchmark for all subsequent models.

1.4 Model Improvement

To improve the full model, backward stepwise selection was applied using both AIC and BIC as stopping criteria. Predictors are removed iteratively starting from the least significant, stopping when further removal increases the information criterion.

Table 7: Coefficient estimates for the full model.

Predictor	Estimate	Std. Error	t-value	p-value	Significant
(Intercept)	37.2890	5.3528	6.9663	0.0000	Yes
crim	-0.1267	0.0355	-3.5692	0.0004	Yes
zn	0.0366	0.0145	2.5171	0.0122	Yes
indus	0.0173	0.0651	0.2655	0.7907	No
chasYes	2.7798	0.9154	3.0367	0.0026	Yes
nox	-15.0930	4.4551	-3.3878	0.0008	Yes
rm	4.2751	0.4576	9.3427	0.0000	Yes
age	-0.0097	0.0143	-0.6734	0.5011	No
dis	-1.2724	0.2128	-5.9781	0.0000	Yes
rad	0.2665	0.0715	3.7263	0.0002	Yes
tax	-0.0128	0.0040	-3.2285	0.0014	Yes
ptratio	-1.0600	0.1387	-7.6417	0.0000	Yes
lstat	-0.4701	0.0567	-8.2974	0.0000	Yes

Table 8: Coefficient estimates for the Backward AIC model

Predictor	Estimate	Std. Error	t-value	p-value	Significant
(Intercept)	37.4586	5.3255	7.0339	0.0000	Yes
crim	-0.1270	0.0354	-3.5857	0.0004	Yes

zn	0.0374	0.0142	2.6241	0.0090	Yes
chasYes	2.7545	0.9067	3.0378	0.0025	Yes
nox	-15.6892	4.1666	-3.7654	0.0002	Yes
rm	4.2066	0.4469	9.4127	0.0000	Yes
dis	-1.2481	0.1998	-6.2452	0.0000	Yes
rad	0.2668	0.0687	3.8852	0.0001	Yes
tax	-0.0125	0.0036	-3.4750	0.0006	Yes
ptratio	-1.0585	0.1373	-7.7100	0.0000	Yes
lstat	-0.4827	0.0524	-9.2151	0.0000	Yes

Table 9: Coefficient estimates for the Backward BIC model. Both AIC and BIC criteria converged on the same reduced model.

Predictor	Estimate	Std. Error	t-value	p-value	Significant
(Intercept)	37.4586	5.3255	7.0339	0.0000	Yes
crim	-0.1270	0.0354	-3.5857	0.0004	Yes
zn	0.0374	0.0142	2.6241	0.0090	Yes
chasYes	2.7545	0.9067	3.0378	0.0025	Yes
nox	-15.6892	4.1666	-3.7654	0.0002	Yes
rm	4.2066	0.4469	9.4127	0.0000	Yes
dis	-1.2481	0.1998	-6.2452	0.0000	Yes
rad	0.2668	0.0687	3.8852	0.0001	Yes
tax	-0.0125	0.0036	-3.4750	0.0006	Yes
ptratio	-1.0585	0.1373	-7.7100	0.0000	Yes
lstat	-0.4827	0.0524	-9.2151	0.0000	Yes

Both the AIC and BIC backward stepwise procedures removed `indus` and `age`, converging on the same 10-predictor model. This agreement between two independent criteria strengthens confidence in the variable selection outcome.

1.4.1 Variable Selection

Table 10: Comparison of full and backward stepwise models by number of predictors, AIC, BIC, 10-fold CV RMSE, and CV R-squared. The best model by CV RMSE is highlighted.

Model	Predictors	AIC	BIC	CV_RMSE	CV_Rsquared
Full	12	2369.02	2424.90	4.7251	0.7550
Backward AIC	**10**	**2365.56**	**2413.46**	**4.6104**	**0.7523**
Backward BIC	10	2365.56	2413.46	4.6716	0.7509

From Table **??**, the Backward AIC model achieves the lowest CV RMSE of 4.6104 against the full model's 4.7251, making it the preferred model among the three. The identical AIC and BIC values for the two backward models confirm they selected the same predictors.

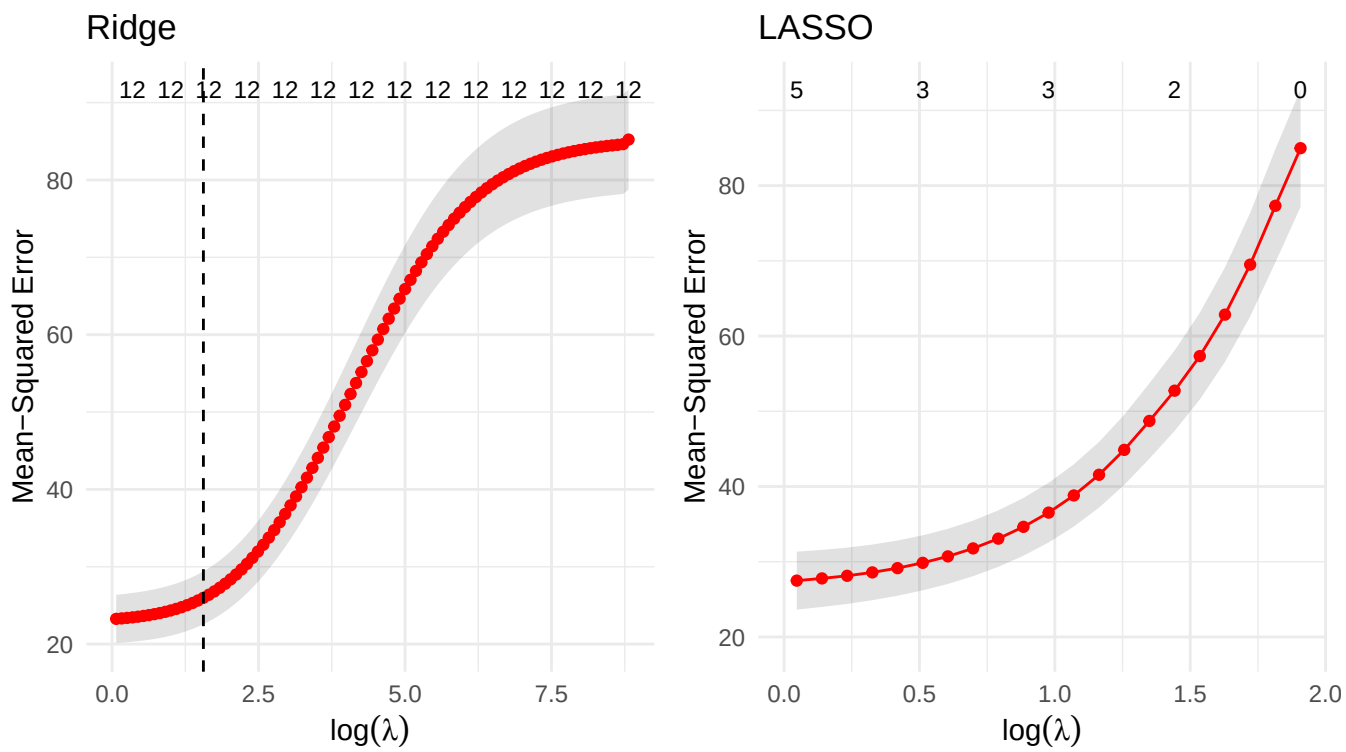

Figure 8: 10-fold cross-validation MSE curves for Ridge (left) and LASSO (right) regularisation

1.4.2 Regularisation

In Figure 8 regularisation was applied to control model complexity and guard against overfitting. Using `lambda.min` — the penalty value minimising 10-fold CV error — Ridge achieved a CV RMSE of 4.8033 and LASSO achieved 4.7831. Ridge retains all 12 predictors throughout the penalty path, shrinking coefficients toward zero without eliminating them. LASSO progressively zeroes out weaker predictors as the penalty increases, performing automatic variable selection.

1.4.2.1 Goodness of Model Fit

Table 11: Final model comparison by 10-fold cross-validated RMSE across all candidate models. The best-performing model is highlighted.

Model	CV_RMSE
Backward AIC	4.6104
Backward BIC	4.6716
Full	4.7251
LASSO	4.7831
Ridge	4.8033

From Table **??**, the Backward AIC model is the best-performing model with a CV RMSE of 4.6104, outperforming all alternatives including regularisation methods. Its CV RMSE is 0.1147 lower than the full model, confirming that the removed predictors contributed noise rather than signal.

The selected interaction term is rm:lstat. `rm` measures the average number of rooms per dwelling, while `lstat` measures the percentage of lower-status residents. The negative `rm:lstat` coefficient indicates that the price premium associated with having more rooms is not constant: extra rooms add more value in lower-`lstat` suburbs than in higher-`lstat` ones, reflecting that spacious homes in affluent areas command a disproportionately higher price.

1.5 Residual Diagnostics

Table 12: Numerical residual diagnostic summary for the final interaction model.

Diagnostic	Value
Mean residual	0.0000
Residual standard deviation	3.7660
Largest absolute standardised residual	7.0787
Observations with |standardised residual| > 2	17.0000

1.6 Conclusion

This analysis set out to predict median home values across Boston suburbs and to identify which suburb characteristics most strongly influence housing prices. Across all modelling approaches evaluated, the Backward AIC model with an rm:lstat interaction term emerged as the best-performing model, achieving the lowest cross-validated RMSE of 4.6104. The retained predictors in the final model point to three broad categories of influence on median home values:

Socioeconomic status is the strongest driver. `lstat`, the percentage of lower-status residents, has the largest negative relationship with `medv`as suburbs with higher lower-status residents have lower median home values.

Property characteristics also matter substantially. `rm`, the average number of rooms per dwelling, has the strongest positive relationship with `medv`. `ptratio`, the pupil-teacher ratio,has a negative effect, suggesting that access to better-resourced schools impacts home prices.

Environmental and structural factors such as nitrogen oxide concentration (`nox`), highway accessibility (`rad`), property tax rate (`tax`), and crime rate (`crim`) all contribute negatively to home values with these houses seeing lower median prices. Distance to employment centres (`dis`) has a positive effect, suggesting suburbs further from industrial areas are valued more highly.

Notably, `indus` and `age` were dropped by both backward selection criteria, confirming that the proportion of non-retail business acres and housing age add no meaningful predictive signal once the other variables are accounted for. The final model performs reasonably well, explaining approximately 59.4% of variation in median home values on unseen suburbs.

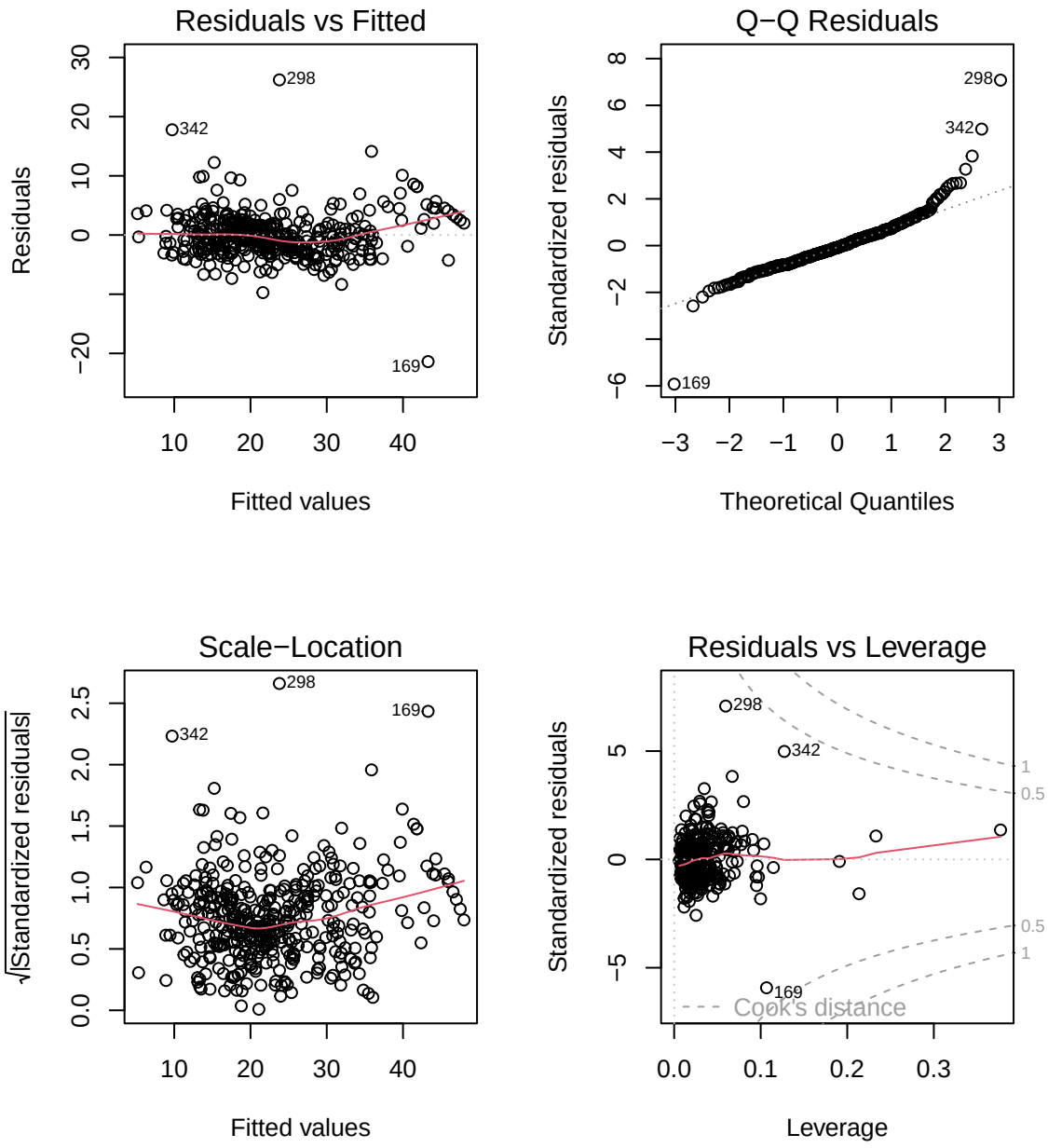

Figure 9: Standard residual diagnostic plots for the final interaction model: (top-left) residuals vs fitted values